AquaPearl 500 Pro

Flagship Smart Multi-Stage RO + UV + UF Water Purifier

Comprehensive User Operation and Engineering Service Manual

Model Code: AP500-PRO-REV3 | Firmware 2.4.0 Compatible

Thank you for selecting Pearlz Home Systems as your healthy living partner. The AquaPearl 500 Pro is engineered with multi-patented precision reverse osmosis, food-grade mineral replenishment, continuous ultraviolet sterilization, and digital IoT telemetry. Please read this manual carefully before installation and commissioning.

Pearlz Home Systems (Pvt.) Ltd. -- Customer Care Desk

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1. Box Contents and Package Inspection

Carefully inspect the shipping carton prior to opening. If the tamper-evident security seal is broken, immediately photograph the carton and contact your delivery carrier. The package must contain all items below:

1x AquaPearl 500 Pro Main Unit	- Integrated chassis with pre-fitted booster pump, PCB, and solenoid bank.
1x Pre-Filter Housing and Cartridge	- Heavy-duty 10-inch transparent spun polypropylene housing with wall bracket.
1x Sediment Cartridge (5-Micron)	- Spun melt-blown fiber cartridge for preliminary grit and particulate capture (SKU-FP-01)
1x Heavy-Duty Brushed Chrome Faucet	- Lead-free stainless steel dispensing faucet with ceramic disc valve mechanism.
5 Meters Food-Grade PE Tubing (1/4 inch)	- High-pressure polyethylene tubing in clean white.
3 Meters Food-Grade PE Tubing (3/8 inch)	- High-volume tubing for cold water inlet connection.
1x Feed Water Diverter Valve (1/2 to 3/8)	- Solid chrome-plated brass angle stop adapter with manual quarter-turn shutoff.
1x Drain Saddle Clamp Assembly	- Engineered wastewater drain clamp fitting standard 40mm sink drain pipe.
1x Filter Spanner Wrench	- Ergonomic circular wrench for servicing pre-filter and membrane canisters.
1x Roll PTFE Thread Seal Tape	- High-density Teflon sealing tape for leak-proof connections.
1x Mounting Hardware Pack	- Stainless steel masonry screws, wall anchors, and drill alignment template.

2. Technical Specifications and Operating Limits

Product Model Name	AquaPearl 500 Pro Smart Water Purifier
Filtration Technology	7-Stage RO + UV-C + UF + Alkaline Mineralizer + Active Carbon
Purification Flow Rate	15 Liters per Hour (75 Gallons per Day) at 25 deg C
Internal Storage Capacity	9.5 Liters Hydrostatic Food-Grade ABS Antimicrobial Tank
Applicable Raw Water TDS	100 PPM up to 2,000 PPM (Total Dissolved Solids)
Purified Output TDS Target	30 PPM to 70 PPM (Post mineral fortification)
Operating Inlet Pressure	15 PSI to 45 PSI (1.0 bar to 3.1 bar)
Maximum Inlet Pressure Limit	50 PSI (Pressure-reducing valve mandatory above 45 PSI)
Feed Water Temperature	5 deg C to 40 deg C (Do NOT connect to hot water source!)
Rated Electrical Power	45 Watts Operating / 1.5 Watts Standby
Input Electrical Voltage	220V - 240V AC, 50Hz (Requires 3-pin grounded outlet)
Dimensions (W x D x H)	380 mm x 240 mm x 480 mm
Net Weight / Gross Weight	8.2 kg empty / 17.7 kg fully filled with water
Wireless Connectivity	2.4 GHz IEEE 802.11 b/g/n (5 GHz not supported)

3. Control Panel and Display Indicators

The AquaPearl 500 Pro features a high-definition capacitive black tempered-glass control console. The display provides real-time water quality metrics, system diagnostics, and filter maintenance alerts.

Power Indicator (Blue Circle)	Solid Blue: System operational. Blinking: System in self-cleaning flush cycle.
TDS Numerical Display	Displays real-time output water purity in Parts Per Million (PPM). Normal: 30-70 PPM.
Filter Life Bar (Multi-color)	Green (>20% life remaining), Yellow (10-20% remaining - order replacements), Blinking Red (<10% - imminent lockout).
UV Sterilization Icon	Glowing Violet: UV lamp active in secondary chamber. Flashing: UV bulb fault (E-04).
Child-Lock Key (Padlock Icon)	Press and hold for 3 seconds to toggle safety lock on dispensing tap.
RESET Button (Touch Key)	Used to clear alert alarms and calibrate newly installed filter cartridges. Press and hold 5 seconds to clear E-03.

4. Operating Instructions and Daily Usage

4.1 Initial Daily Dispensing: Upon waking, dispense and discard the first 200 ml of purified water to clear standing water from the faucet spout.

4.2 Continuous Filtration: The purifier operates automatically. When tank water level falls below 75%, the internal low-noise booster pump engages and purifies fresh water until the upper float switch triggers cut-off.

4.3 Vacation Mode: If leaving the residence for more than 48 hours, close the blue angle feed valve, unplug the AC adapter, and drain the storage tank via the bottom drain port.

5. Filter Replacement Intervals and Life Cycle

Filter longevity depends on incoming raw water particulate density and total household volume. The table below indicates standard replacement intervals for water with 500 PPM feed TDS:

Stage 1: Sediment Pre-Filter	**SKU-FP-01**	**3 to 6 Months**	**Every 2,500 Liters**	Protects booster pump and membrane from sar
Stage 2: Activated Carbon Block	**SKU-AC-03**	**6 Months**	**Every 4,000 Liters**	Absorbs free chlorine, organic pesticides, chem
Stage 3: RO Membrane Element	**SKU-RO-02**	**24 to 36 Months**	**Every 15,000 Liters**	Eliminates arsenic, lead, fluorides, heavy metal
Stage 4: Post-Carbon Mineralizer	**SKU-MN-04**	**12 Months**	**Every 6,000 Liters**	Adds micro-doses of Magnesium (Mg^{2+}) and C

6. Cleaning, Descaling, and Tank Sanitization

6.1 Quarterly Surface Cleaning: Wipe the exterior high-gloss piano black housing with a soft microfiber cloth dampened with mild soapy water. Never apply benzene, thinner, or abrasive cleaning powders.

6.2 Bi-Annual Descaling Procedure: In areas with high calcium hardness (>300 PPM CaCO3), the internal heating elements and RO membrane sheath should be descaled every six months:

1. Turn off main cold inlet valve and empty water tank.
2. Dissolve 50 grams of food-grade anhydrous citric acid in 2 liters of warm water (approx 40 deg C).
3. Introduce descaling solution into the pre-filter housing bowl.
4. Power on unit and allow solution to circulate through internal manifolds for 20 minutes.
5. Flush system thoroughly with 15 liters of pure tap water until effluent tests neutral on litmus paper.

7. Safety Warnings and Operating Precautions

- DANGER: Electrical Shock Hazard. Never touch the power adapter with wet hands.
- DANGER: Hot Water Contamination. Never connect feed line to hot water pipe (>45 deg C). Doing so melts RO thin-film composite membrane.
- WARNING: Grounding Mandatory. The power supply must be connected to an authenticated three-pin grounded socket.
- CAUTION: Cabinet Protection. Always check tubing quick-connect locking clips during installation. Pearlz assumes no liability for cabinet water damage.
- CAUTION: Unauthorised Repair. Opening internal circuit housing by uncertified persons permanently revokes product warranty.

8. Audible Alert Codes and Meaning (E-01 to E-08)

E-01	**1 Long Continuous Beep**	**Low Feed Water Pressure (<15 PSI)**	Verify main house water valve is fully opened; inspect pre-filter for n
E-02	**2 Short Beeps (repeating)**	**Booster Pump Thermal Overheat (>65 deg C)**	System enters 30-min auto cool-down. Verify ventilation around unit
E-03	**3 Short Beeps + Red Flash**	**Sediment Pre-Filter Saturation**	Replace sediment filter cartridge (SKU-FP-01). Hold RESET button
E-04	**4 Short Beeps + Violet Flash**	**UV-C Sterilization Module Fault**	UV lamp expired or driver failed. Call Pearlz service van for replace
E-05	**Continuous Rapid Siren**	**Internal Base Flood Leak Detected**	EMERGENCY: Shut feed valve immediately! Drip tray float sensor t
E-06	**Intermittent Double Chime**	**High Purified Water TDS (>150 PPM)**	RO membrane failure or TDS bypass valve open. Inspect membran
E-07	**1 Short Beep every 60 sec**	**Storage Tank Float Sensor Jammed**	Tank sensor stuck. Tap top tank cover gently; check internal float w
E-08	**Rapid High-Pitch Staccato**	**AC Mains Voltage Irregular (<180V / >260V)**	Electronic voltage protector engaged. Connect via 1KVA automatic

9. In-Field Service Diagnostics and Flow Control

When diagnosing hydraulic issues in the field, technicians must verify:

1. Feed pressure at point-of-entry using oil-filled 0-100 PSI pressure gauge.

2. Reject-to-Pure Water Ratio: Factory calibrated ratio is approximately 1.5:1 (1.5 liters reject for every 1 liter pure).

3. Auto-Shutoff Valve (ASV) Seating: When dispensing faucet is closed, waste water flow must cease within 90 seconds.

10. Hydraulic and Electronic Schematic Overview

The hydraulic path proceeds sequentially through:

1. Raw Inlet (3/8 inch PE) -> Low Pressure Switch (LPS) -> 24V DC Inlet Solenoid Valve (SV1).
2. 5-Micron Spun Sediment Pre-Filter Bowl -> Extruded Carbon Block.
3. High-Pressure 24V Diaphragm Booster Pump (110 PSI maximum boost).
4. 75 GPD Thin-Film Composite Reverse Osmosis (TFC RO) Vessel.
5. Permeate line -> High Pressure Switch (HPS) -> Post-Carbon and Calcite Mineralizer -> UV Chamber -> Storage Tank.
6. Brine line -> Auto-Flush Solenoid (SV2) -> 450cc Capillary Flow Restrictor (FR) -> Drain Saddle.

11. Customer Support and Warranty Service Centers

To register your 24-month warranty, visit https://pearlzhome.example/register or scan the QR code on your warranty card within 14 days of purchase. Keep your original sales receipt.

Official Contact Channels:
- Central Support Email: support@pearlzhome.example
- Warranty Claims Desk: claims@pearlzhome.example
- WhatsApp Support Desk: +92-42-111-PEARLZ (08:00 - 24:00 PKT)
- Corporate Headquarters: Pearlz Home Systems (Pvt.) Ltd., Gulberg III, Lahore, Pakistan.

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